Def. A injective regular mapping $X : D \to \mathbb{R}^3$ on open $D \subseteq \mathbb{R}^2$ is called a patch.

Def. A subset $M \subseteq \mathbb{R}^3$ is called a surface if:

for each $p \in M$, there exists a patch $X : D \to \mathbb{R}^3$ such that $B_r(p) \cap M \subseteq X[D]$ for some $r > 0$.